

Trung Pham

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Data Engineer, 4 years of production pipelines on GCP and AWS, where cost, reliability, and observability are measured rather than asserted — 42% lower ingestion cost, 36% lower storage cost. The same discipline carried into three public end-to-end ML pipelines, built so their evaluation numbers are defensible.

CORE COMPETENCIES

- **Data engineering (GCP & AWS):** Dataproc/PySpark, BigQuery, Cloud Functions, GCP Workflows, Cloud Storage; AWS — Step Functions, Lambda/Glue, S3/Athena, DynamoDB, Iceberg; Terraform & CDK (IaC); Docker; ETL, data-lake design & migration; SQL optimization.
- **Reliability, observability & cost (FinOps):** Prometheus/StatsD, SLI/SLO-as-code; GitLab CI/CD.
- **ML / MLOps & languages:** PyTorch, MLflow tracking + model registry, Medallion Parquet layers with Pandera schema gates, pytest/CI, FastAPI + Docker serving; Python, SQL, Git, Linux/WSL2.

PROFESSIONAL EXPERIENCE

Data Engineer 2024–Present
Semrush, Remote

- Primary developer of a large-scale web-crawl pipeline on GCP, end-to-end: Terraform IaC (multi-module, prod/staging), Dataproc PySpark ETL, Cloud Functions, GCP Workflows orchestration, and a partitioned BigQuery model over landing → staging → production Parquet tiers.
- Own reliability and observability: shared retry logic, concurrent-DML handling, staging-table MERGE patterns; StatsD/Prometheus metrics, SLI/SLO-as-code, failed-workflow alerting, BigQuery audit tables; schema governance (managed → external tables).

Data Engineer 2022–2024
RYTE Technology Services

- Designed a serverless AWS batch platform: Step Functions over Lambda — Glue as the OOM fallback — Parquet on S3 with Athena, EventBridge chaining, DynamoDB audit table, Apache Iceberg transactional serving; infrastructure as code in AWS CDK (TypeScript).
- Built and migrated the central data lake (~5 GB/day) onto it: ingestion cost —42%, storage cost —36%.

Research Intern 2020
National Chung Cheng University (CCU), Taiwan — MOE-funded scholarship; 3D pose estimation for patient monitoring

SELECTED PROJECTS

WM-811K Wafer-Defect Classification github.com/pqtrng/wm811k-defect-detection
PyTorch ResNet-18 on 64×64 wafer maps, macro-F1 0.828 → 0.920; Medallion Parquet layers with Pandera gates, MLflow model registry, pytest CI, FastAPI/Docker serving with Grad-CAM.

Cyber-Physical / OT Anomaly Detection (HAI ICS) github.com/pqtrng/cps-anomaly-pipeline
Leakage-safe LSTM-autoencoder pipeline on industrial control-system telemetry: raw F1 0.792 vs a strong z-score baseline; 25/25 attacks detected, 4-second median latency, 88% within a 30-second SLO.

CircuitNet-N14 Routing-Congestion Prediction (EDA) github.com/pqtrng/circuitnet-congestion
Evaluation-methodology finding on 14nm place-and-route feature maps: the checkpoint-selection rule is a confound of the same order as the loss function, and the selection gap held when checkpoints were scored once on the held-out test design — over leakage-safe design-wise splits.

EDUCATION

BSc in Computer Science — Vietnamese-German University (VGU) 2016–2020
In dual-degree partnership with Frankfurt University of Applied Sciences. GPA: 1.6 (German scale, ≈ 8.6/10).

LANGUAGES

Vietnamese (native), English (IELTS 6.5), Mandarin Chinese (TOCFL A2; exam scheduled Dec 2026).